

An Exploratory Investigation into the Design of a Basketball Immersive Vision Training System

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ABSTRACT

Having good vision ability is important for basketball players who possesses the ball to efficiently search the wide-open teammates and quickly pass the ball to the one who has better chance to score according to the defenders' movements. To customize precise training scenarios for cultivating an individual athlete's vision abilities, we proposed a basketball immersive vision training system which considers not only the real-world vision training scenario in basketball team but also the concept of the optimal gaze behaviors to provide more reliable training tasks. Moreover, a four-week pilot study is also designed to evaluate whether the proposed system can train the vision ability of basketball players. The result of this pilot study shows the recruited five participants satisfied the experience of proposed training system.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Virtual reality—; Applied computing—Education—

1 INTRODUCTION

Basketball is a five-on-five team ball game designed to score as many points as possible in a limited time to win the game. In each basketball play, the player possessing the ball needs to have a good vision ability to efficiently search the wide-open teammates and quickly pass the ball to the one who has better chance to score according to the defenders' movements. The vision behavior of the ball handler is different when he/she is in the stop-ball state and in the dribbling state. In the stop-ball state, the ball handler has 5 seconds to observe whom is a better candidate to pass the ball. In the dribbling state, the ball handler has much shorter time to observe the situation and make the passing decision. Moreover, the ball handler needs to dribble smoothly and not drop the ball during dribbling. To sum up, developing basketball players' passing vision includes improving the ability to search wide-open opportunities and observe defenders around wide-open opportunities, as well as maintaining the stability of the dribble to prevent turnover.

In our previous work [4], we conduct semi-structured interviews with three basketball coaches of college basketball teams to understand their needs and the way they have achieved vision training. According to the interview, there are currently no universal player vision training methods. One of the interviewed coaches introduced

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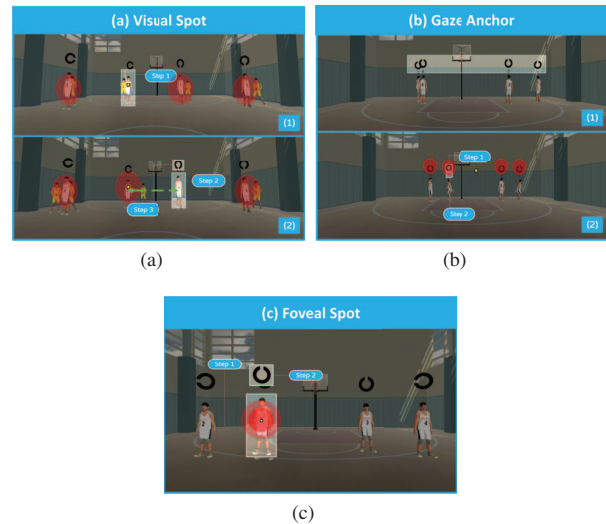


Figure 1: Example of training task for different optimal gaze behaviors (a) visual spot (b) gaze anchor (c) foveal spot

two vision training scenarios he designed to train his players. One of the training scenarios is four players are asked to stand in their attacking positions, and each player makes a gesture indicating a number randomly. The trainee has to calculate the sum of all gesture numbers and answer the summation quickly. The coach mentioned that this kind of training scenario aims to not only speed up the trainee's reaction time when he/she sees an opportunity but also improve the trainee's cognitive aspects of mental calculation, which enforce the trainee to process more information about the current situation. Inspired by above-mentioned scenario, we design a number accumulation task for the trainee practices in VR, the proposed system generates and shows random numbers from 1 to 10 above the heads of virtual players. The trainee has to scan the entire court to accumulate all these numbers and respond the answer as soon as possible. In this way, it can accurately replicate the training content as well as efficiently digitalize the training process, resulting in lower labor and time costs for training players' vision abilities.

In this work, we aim to examine the effectiveness of the proposed basketball immersive vision training system. Therefore, we design a four-week study procedure and recruit five participants as trainees to evaluate whether the proposed system can train the vision ability of basketball players effectively.

2 TRAINING TASK DESIGN

To ensure the training task can facilitate vision training, we further consider the optimal gaze behaviors proposed by Klostermann *et*

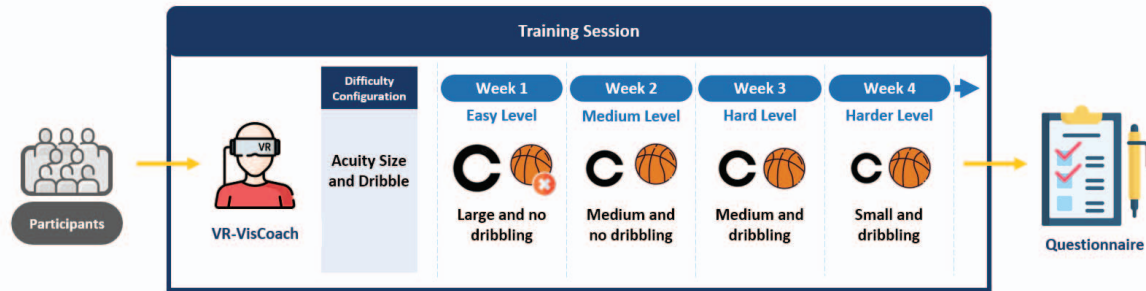


Figure 2: Study procedure

al. [3] to design a comprehensive training task. Klostermann *et al.* divided the functions of the eyes into foveal vision which acquires detailed information, and peripheral vision, which is sensitive to object changes. They proposed three optimized gaze behaviors depending on the complementarity between the two visual abilities as follows. (a) **Visual Spot:** Athletes not only need to fixate on an object but also need to use peripheral attention to select the next gaze position to switch to. (b) **Gaze Anchor:** Athletes gaze in free space and process information through peripheral vision. (c) **Foveal Spot:** Athletes process information about their current gaze position through foveal vision. We integrate the idea of optimal gaze behaviors with the proposed system to design three training tasks shown in Figure 1.

3 STUDY PROCEDURE DESIGN

We design a four-week controlled study to evaluate whether the proposed system can train the vision ability of basketball players effectively. As shown in Figure 2, the training session last for four weeks, and the difficulty of the tasks increase steadily every week. Each participant is asked to wear a StarVR HMD (with higher horizontal FoV and much wide vertical FoV than other commercial HMDs) for experiencing the training task. A microphone and a Leap Motion sensor are attached to the HMD, so the trainee can respond the answer and interact with the system via voice and gestures. Moreover, a real basketball is used for dribbling during the training. After a four-week controlled study, each participant is asked to fill in the questionnaires.

4 RESULT OF PILOT STUDY

we recruited five participants from school and department basketball team, among which four were female, and one was male. Two participants reported having a VR experience, and three others had never experienced VR. We collected subjective responses to the proposed VR vision training system through a qualitative survey. The questionnaire covers the following four aspects.

- **Usability.** We used a System Usability Scale (SUS) [1] with five-point Likert scale to explore the score of usability, in which a higher score indicates better usability. The result of usability in the SUS questionnaire was Mean = 89 and SD = 4.54, indicating the participants gave positive feedback on how to use the proposed immersive training system.
- **Sports Imagery Ability.** Sports imagery ability questionnaire proposed by Williams *et al.* [6] was used to explore whether the participants felt the skill imagery and the strategic imagery of improving their vision ability. The results showed that it obtained 5.8 out of seven-point Likert scale for the *Skills Imagery* term, and 6 out of seven-point Likert scale for the

Strategy Imagery term, indicating the proposed system can improve their sports imagery ability.

- **Presence.** We applied the Igroup Presence Questionnaire (IPQ) [5] to explore the level of immersion on a seven-point scale. The result indicated that the participants can provide the immersive environment easily.
- **Simulator Sickness.** Kennedy Lane Simulator Illness Questionnaire [2] was used to measure the participants' discomfort with the proposed system. We found that *sweating* was caused by the fact that participants had to dribble during the task while experiencing the proposed immersive training system. The HMD may wobble slightly when the participant was dribbling, resulting in *difficulty focusing*.

5 CONCLUSION AND FUTURE WORK

In this paper, a novel VR vision training system for basketball players is proposed to simulate the player's viewing perspective on the court as well as generate the designed vision training tasks. The results show that the participant satisfied the proposed vision training system after a four-week training session. In the future, we will design a metrics to evaluate the reaction time and correctness before and after experiencing the training session. Also, we will try to compare with the conventional vision training system to understand the effectiveness of the proposed system.

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